

# INTE 42312 – Virtual and Augmented Reality

## Individual Assignment (25% Continuous Assessment)

Title: Design and Development of an Interactive WebXR Experience

### 1. Background

Extended Reality (XR) technologies are increasingly delivered through web browsers using WebXR and WebAR technologies. This assignment provides students with an opportunity to design and develop an interactive browser-based XR experience that demonstrates core augmented reality concepts while applying software engineering principles.

### 2. Assignment Objective

Individually design, implement and evaluate an interactive WebXR/WebAR experience that addresses a real-world use case. The emphasis is on demonstrating technical competency, user interaction design and appropriate application of XR technologies rather than developing a commercially complete product.

### 3. Learning Outcomes

- Apply fundamental concepts of augmented and extended reality.
- Develop an interactive WebXR/WebAR experience using suitable frameworks.
- Implement meaningful user interaction with 3D digital content.
- Evaluate usability and technical implementation of an XR solution.

### 4. Technical Requirements

#### Core Requirements:

- Browser-based solution (mobile compatible).
- Use an approved framework (Three.js, A-Frame, AR.js, Babylon.js, or Model Viewer).
- Include at least two 3D models, appropriately optimized for web delivery (polygon reduction, texture compression, format conversion). Original modelling is **not** expected.
- Include a user interface for XR/AR and appropriate animation, lighting, or audio.

#### Tracking Requirements (Both Must Be Demonstrated):

- **Marker-Based AR:** Implement tracking using a specific image, marker (e.g., Hiro), or QR code to anchor digital content.
- **Markerless AR:** Implement spatial tracking (e.g., plane detection, hit-testing, or environment mapping) using the WebXR Device API or supported browser equivalents.

### **Advanced Implementation (Choose At Least One):**

In addition to the tracking requirements above, the markerless experience must incorporate at least one of the following:

- **Option A (Data-Driven):** Integrate live data from an external REST API or database to dynamically alter the 3D content or environment (e.g., visualizing live precision agriculture sensor data or intelligent transport system telematics).
- **Option B (Complex Interaction):** Provide advanced user interactions beyond basic clicks (e.g., multi-step state management, drag-and-drop object manipulation, or physical collision detection between 3D elements).

## **5. Suggested Application Domains**

Below are some domains suggested for the Problem solving application. However it is not limited to as students are encouraged to pick a domain of choice and interest.

- Education
- Healthcare
- Tourism
- Retail
- Manufacturing
- Telecommunications
- Agriculture
- Cultural Heritage
- Entertainment
- Smart Cities & Intelligent Transport Systems

## **6. Submission Requirements**

| <u>Deliverable</u>        | <u>Requirement</u>                                                                                                                              |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Source Code</b>        | Complete project files (e.g., Git repository link or zipped directory)                                                                          |
| <b>Hosted Application</b> | Public URL (GitHub Pages, Netlify, Vercel, etc.)                                                                                                |
| <b>Demonstration</b>      | 3 minute demonstration covering all required features                                                                                           |
| <b>Technical Report</b>   | Maximum 4 pages describing the problem, design, implementation, testing, and reflection.<br><b>This report MUST include a dedicated section</b> |

| <u>Deliverable</u> | <u>Requirement</u>                                                                                                               |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------|
|                    | <b>detailing specific technical challenges faced during development and the exact strategies/solutions used to rectify them.</b> |

## 7. Assessment Criteria - (25 Marks)

Submissions will be evaluated based on the following core pillars:

- **Technical Implementation:** Successful integration of the chosen framework, 3D assets, and reliable execution of both marker-based and markerless tracking. **(8 Marks)**
- **User Experience & Interaction:** Smoothness of the interface, usability of the advanced feature (data integration or complex interaction), and overall user journey. **(5 Marks)**
- **Documentation & Troubleshooting:** The depth and clarity of the technical report, particularly the articulation of development challenges and the exact engineering strategies used to resolve them. **(5 Marks)**
- **Problem Definition & Innovation:** Clear articulation of the chosen real-world use case, creativity of the final experience, and justification for why WebXR is an appropriate solution. **(4 Marks)**
- **Testing & Evaluation:** Evidence of proper testing across devices and a clear presentation/demonstration of the final build. **(3 Marks)**

## 8. Academic Integrity

Students may utilise openly licensed platforms, assets and libraries with appropriate acknowledgement. The design, implementation and report must represent the student's own work.

Deadline: **29th September 2026**